

Introduction:-

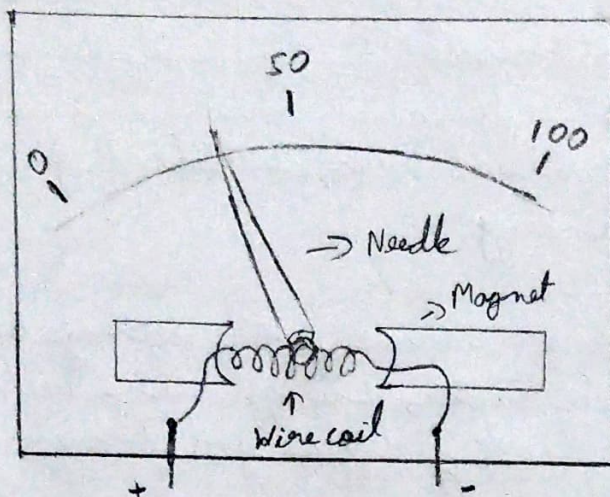
Aim: Introduction to various Basic Instruments of Electrical Technology.

Objective: Introduction to various supply systems, Ammeter, Voltmeter, Wattmeter, Energy meter, Tachometer, Rheostat, loading devices and Transformer.

Apparatus Required: Ammeter, Voltmeter, Wattmeter, Energy meter, Tachometer, Rheostat, Various capacitors, Various Resistors and AC & DC Power supply

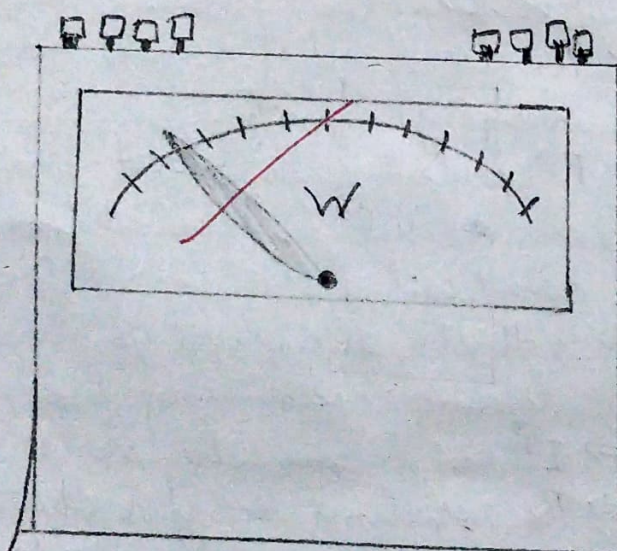
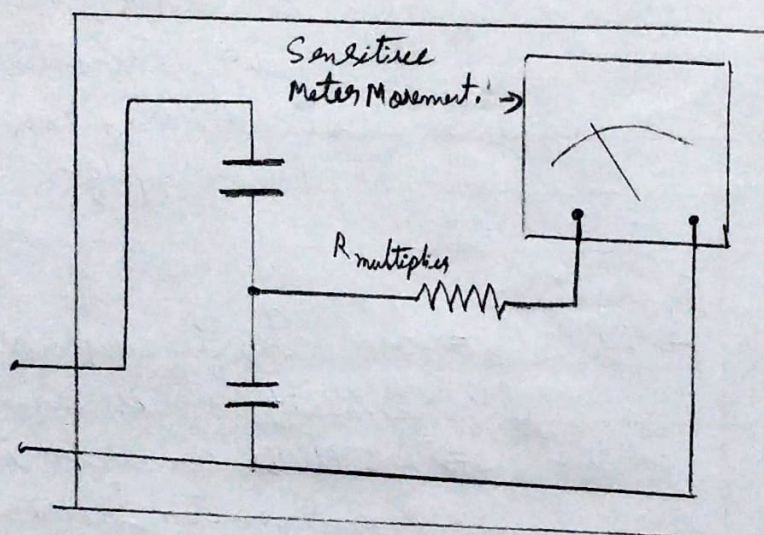
Theory:

- **Ammeter** ⇒ Ammeter is employed to measure current in a circuit and is connected in series.
 - ⇒ As it is connected in series voltage drop across the ammeter terminal is very low.
 - ⇒ Due to which resistance should be as low as possible.
 - ⇒ The current coil of ammeter has low current carrying capacity.
 - ⇒ For protecting the ^{coil} ~~current~~ from high current a low resistance is connected in parallel to the current coil, It is called Shunt Resistance.
- **Voltmeter** ⇒ Voltmeter is employed to measure the potential difference across any two points of a circuit.
 - ⇒ It can be connected across any element ^{circuit} in the circuit.
 - ⇒ The resistance ~~of~~ of voltmeter is kept very high by connecting a high resistance in series of the voltmeter with the current coil.



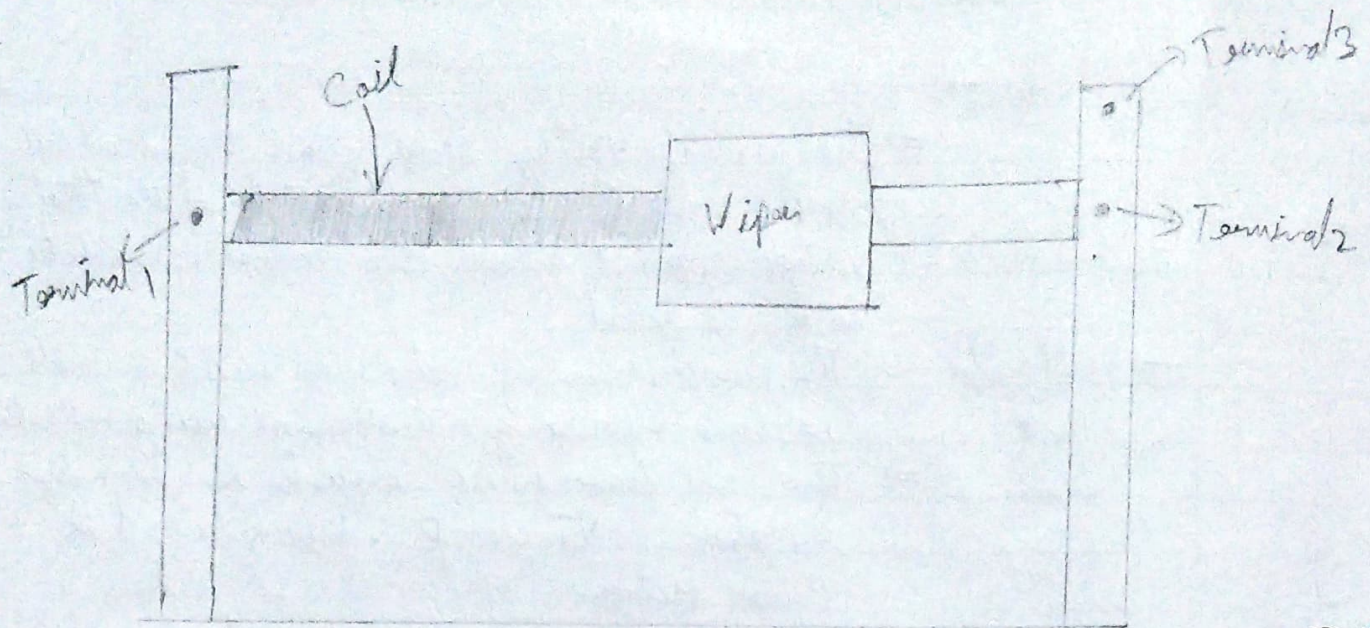
Ammeter

Voltmeter



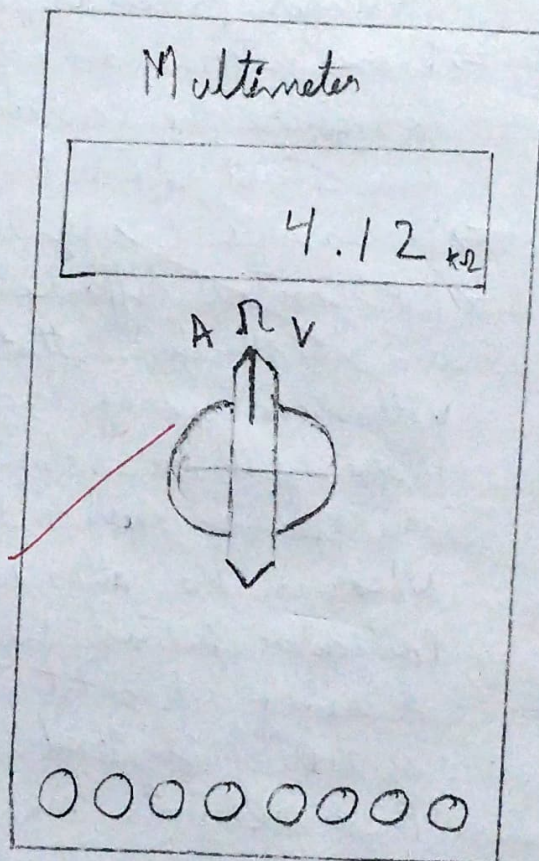
Wattmeter

- ⇒ The actual voltage drop across the current coil of the voltmeter is only a fraction of the total voltage applied across the voltmeter which is to be measured.
- Wattmeter ⇒ The measurement of real power in AC circuits is done by using an instrument using Wattmeter.
- ⇒ The real power in AC circuits is given by expression $VI \cos \phi$, where, $\cos \phi$ is power factor.
- ⇒ A Wattmeter has two coils, namely, current coil & pressure coil. The current coil (CC) is connected in series with the load and pressure coil (PC) is connected across the load.
- ⇒ Wattmeters are available in dual range for voltages as well as for current.
- Energy meter ⇒ Energy meter is an instrument which is used to measure the consumption of electric energy in a circuit (AC or DC). ↑
- ⇒ It measures energy in kWh.
- ⇒ The essential difference between an energy meter and a wattmeter is that the former is fitted with some type of registration mechanism where by all the instantaneous readings of power are summed over a definite period of time whereas the latter indicates the value at particular instant when it is read.
- Tachometer ⇒ A locking rheostat type of ~~cos~~ load consists of no. of identical resistive elements. These elements are connected in series or parallel. The rheostat is made up of high resistivity material such as like ~~nickel~~ nickel.



R hoostat.

Multimeter



Chromium. The elements of the load can be designed to take 1A, 2A or 4A of current.

→ Rheostat ⇒ A rheostat is a variable resistor which is used to control current. &

⇒ They are able to vary the resistance in a circuit without interruption.

⇒ The first connection is made to one end of the resistive element and the other to the wiper.

→ Multimeter ⇒ Multimeter is a measuring instrument used to measure the current, voltage and resistance.

These can be used to troubleshoot many electrical equipments such as domestic appliances, power supplies etc.

→ Transformer ⇒ A transformer is a static device which consists of two or more stationary electric circuits interlinked by a common magnetic circuit for the purpose of transferring electrical energy b/w them.

⇒ The transfer of electric energy takes place from one circuit to another circuit without change in frequency.

⇒ A transformer may be for stepping up voltages from low to high or stepping down voltage from high to low. 28/11

→ Various Supplies ⇒ A.C. Supply systems:

1. Single phase 230V - In this system we have two wires, one is known as phase/line and other is neutral.

2. Three phase 400V - In this we have 3 wires one for each phase or line.

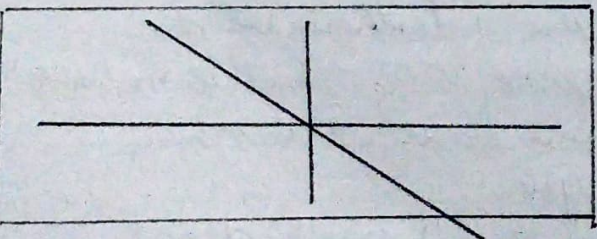
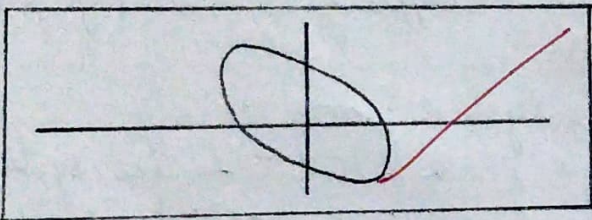
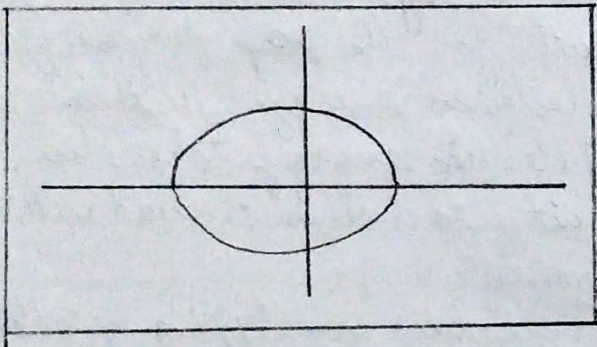
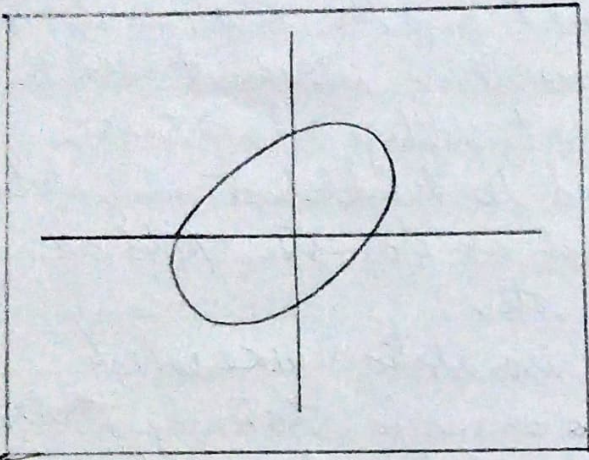
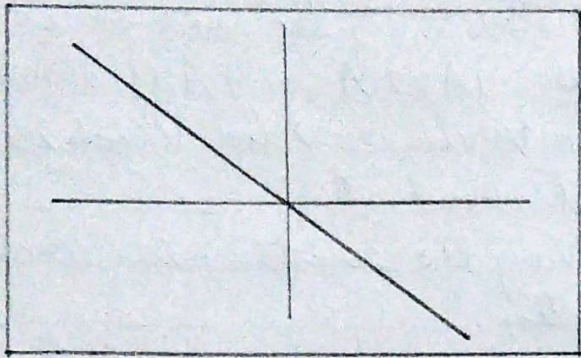
D.C. Supply system

1. From battery - We use rectifiers for 6V or 12V

2. From generator.

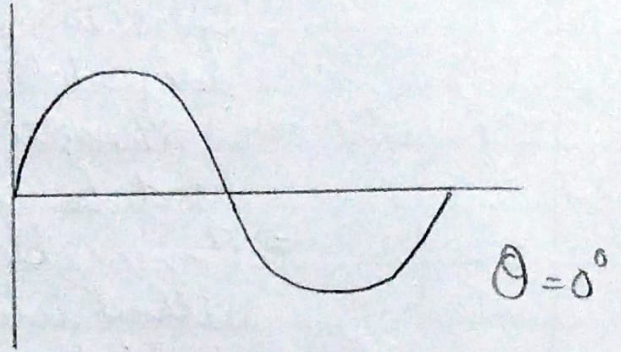
Teacher's Sign

Lissajous figures

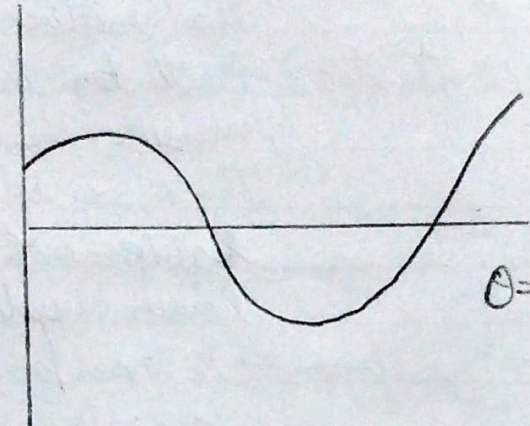


Vertical deflection voltage

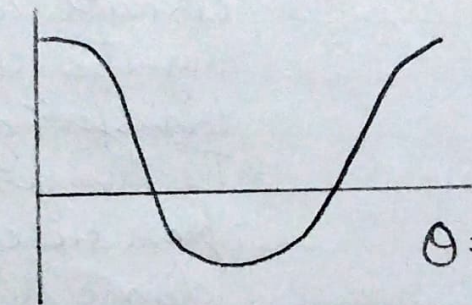
$$e_v = E_m (\sin(\theta + \phi))$$



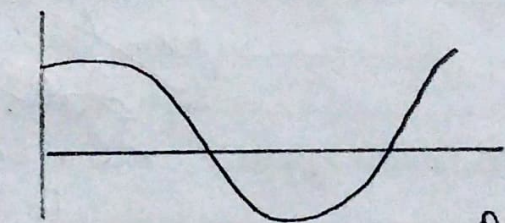
$\theta = 0^\circ$



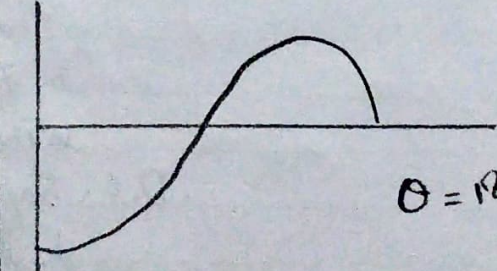
$\theta = 30^\circ$



$\theta = 90^\circ$



$\theta = 150^\circ$



$\theta = 180^\circ$

Experiment : 1

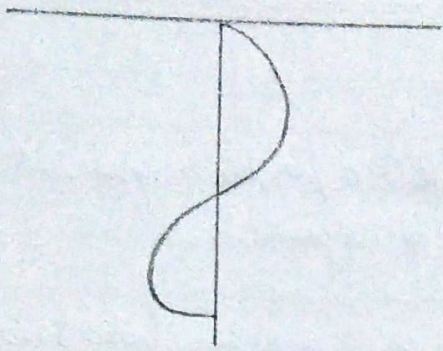
Aim: To study the applications of CRO for the measurement of voltages, frequency and phase of signals.

Theory: An oscilloscope is an electronic measuring device which provides a two dimensional visual representation of a signal. Because the oscilloscope allows the user to see the signal(s), their characteristics can be easily measured and observed. The oscilloscope displays the graph of voltage (on the vertical axis) versus time (on the horizontal axis). Most electrical circuits can be easily connected to the oscilloscope typically with probes. Oscilloscope, previously called an oscillograph and informally known as scope, CRO or DSO is a type of electronic test instrument that allows observation of constantly varying signal voltages, usually as a time-dimensional graph of one or more electrical potential differences using the vertical or 'Y' axis, plotted as a function of time (horizontal or 'X' axis).

Procedure:

Various functions of CRO - Carefully look at the front panel of the CRO and try to understand the various control knobs like vertical gain switch (Volts/div), the horizontal sweep speed selector (Time/div), CAL, Var switch, Entensity and Horizontal shifting knobs X and Y inputs, chop, X10 component test etc.

Voltage Measurement: The ac signal generator is used fed to the Y-input the CRO in the ac mode and the peak to peak voltage of the signal is measured by noting the height of the signal on the screen & the vertical gain position of the input.



Horizontal
deflection
voltage

$$O = em \sin O$$

Amplitude:

$$\text{Voltage/division} \Rightarrow 1V/\text{div}$$

$$\begin{aligned}\text{Amplitude} &= \text{No of blocks} \times \text{Volt per division} \\ &= 3.4 \times 1 \\ &= 3.4 \text{ Volts}\end{aligned}$$

frequency:

$$\text{time/div} = 1 \text{ millisecc/div}$$

$$\text{Time} = \frac{\text{No of block} \times \text{time}}{\text{div}}$$

$$= \frac{1}{1000} \text{ sec}$$

$$= 10^{-3} \text{ sec}$$

$$f = \frac{1}{T} = \frac{1}{10^{-3}} = 1 \text{ KHz}$$

Frequency Measurement: Connect the signal from the function generator (FG) to the Y-input & adjust the horizontal sweep selector (time/div) to get a steady pattern of the signal on the CRO screen. Measure the time interval b/w two peaks, say t sec. Hence determine the frequency of the signal $f = 1/t$.

Phase Measurement: One can approximately measure the phase diff b/w two signals (same frequency) by reading the signal of two input Y_1 & Y_2 of a dual trace CRO & noting the shift in the peak position. The shift is measured on the time scale (div/sec) & then convert into the phase diff assuming one period $T = 2\pi$ radians or 360° .

Atulima
5/12

EXPERIMENT: 2a

Aim: To verify Thevenin Theorem and find out Thevenin's Equivalent circuit using DC sources.

Objective: To apply the principle of Thevenin Theorem for electrical network containing independent DC sources and to find out Thevenin's eqi. circuit.

Apparatus: Digital multimeter, power supply, Resistance (Wire wound) & connecting wires.

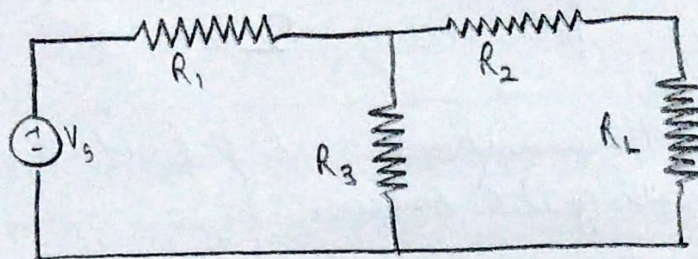
Theory: Sometimes, we wish to determine the response in a single load resistance in a network. Thevenin Theorem enables us to replace the remainder of the network by a single simple equivalent circuit. Determining the response in the load resistance then becomes easier. The use of Thevenin theorem is specially very helpful and time saving when we wish to find the response for different values of load resistance.

Thevenin Theorem states that current through a load resistance connected across any two points of an active network can be obtained by the formula:

$$I_L = \frac{V_{TH}}{R_{TH} + R_L}$$

where ~~V_{TH}~~ V_{TH} is the open circuit voltage at the terminals of R_L when R_L is disconnected and R_{TH} is the equivalent resistance viewed from the output terminals where all the sources replace by their internal resistance only (short circuit the voltage sources)

Circuit Diagram -



Observation Table -

dc voltage (V)	circuit resistance (K Ω)	obs. current (mA)	thevenin's eqn. circuit parameters	cal. current	% error
	R_1, R_2, R_3, R_L	I_L	V_{Th} V_m R_{Th} R_m (obs) (cal) (obs) (cal)	I'_L (mA)	
5V	1 1 1 1	1	2.5 2.5 1.5 1.5	1	0



- Procedure:
1. Build a circuit on the breadboard using resistances R_1, R_2, R_3 & R_L . Adjust the voltage of DC voltage source. Connect an ammeter in series with R_L to measure current.
 2. Switch on the circuit and note the readings on Ammeter.
 3. Remove R_L and measure V_{th} using the digital multi-meter.
 4. Remove the DC voltage source by short-circuiting them (replacing them with a wire). Set the multi-meter to measure the resistance R_{th} .
 5. Draw the Thevenin's equivalent circuit and calculate the load current. It should be equal to the current value obtained in step 2.

Calculations: The calculation of I_L' can be done using the formula $I_L' = \frac{V_{th}}{R_{th} + R_L}$

The percentage error can be calculated as:-

$$\frac{\text{observed value} - \text{calculated value}}{\text{calculated}} \times 100$$

Result: The percentage error was found to be 0%.

Conclusion: The Thevenin theorem is verified and the Thevenin's equivalent circuit is obtained.

EXPERIMENT - 26

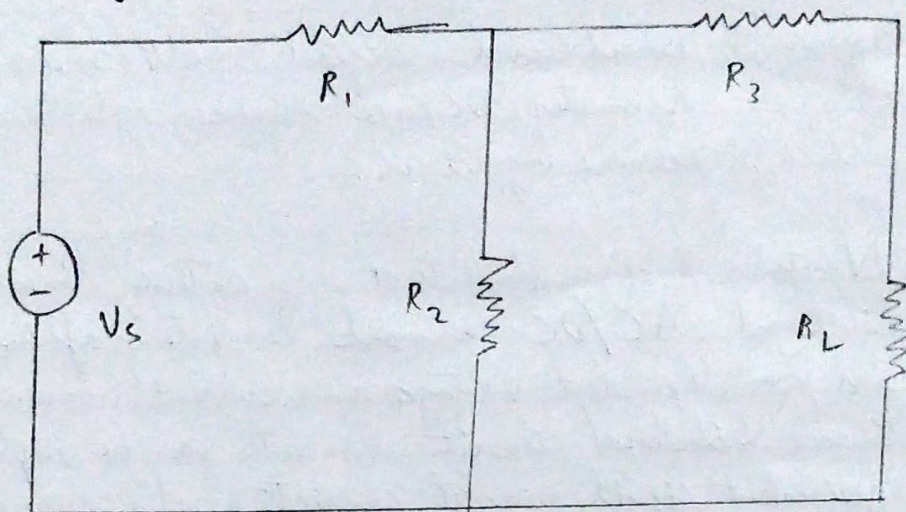
Aim: To verify Norton's theorem and find out the Norton's equivalent circuit using DC sources.

Apparatus Required: Bread board, Digital multi-meter, Digital Ammeter, DC power source, resistances & connecting wires.

Theory: Norton's Theorem states that any active linear ~~bilateral~~ bilateral AC/DC network consisting of independent and/or dependent voltage and current sources and linear bilateral circuit elements can be replaced by an equivalent circuit consisting of a current source and a shunt resistance, the current source being the short-circuit ~~current~~ across the short-circuited load terminals and the resistance being the internal resistance of the source network looking through the open-circuited load terminals. The load current can then be obtained from the Norton's equivalent circuit using the formula
$$I_L = \frac{R_N}{R_N + R_L} \times I_{sc}$$

Where I_{sc} is short-circuited current through the short-circuited link b/w the terminals of load resistance R_L and R_N is the equivalent resistance viewed from the load terminals when all the sources replaced by their internal resistance only, i.e. all voltage sources are short-circuited and the current sources are open circuited.

Circuit Diagram-



DC Voltage Source (V)	Circuit Resistance				Obsd. Current (I_L)	Norton's eq. circuit parameters		Calc. Current (I_L)	% Error
	R_1	R_2	R_3	R_L		$I_{sc} (mA)$	$R_N (k\Omega)$		
5V	0.00059	0.00059	0.00059	0.00268	0.72	0.7	0.836	0.697	3.29

- Procedure:
1. Build a circuit on the bread board using resistances R_1, R_2, R_3 & R_L . Adjust the voltage of DC Voltage source. Connect an ammeter in series with R_L to measure current I_L .
 2. Switch on the circuit and note the readings on Ammeter.
 3. Remove R_L and measure V_N using the digital multimeter.
 4. Remove the DC voltage source by short-circuiting them (replacing them with a wire). Set the multimeter to measure the resistance R_N .
 5. Draw the Norton's equivalent circuit and calculate the load current. It should be equal to the current value obtained in Step 2.

Calculations: The calculation for I_L' can be done using the formula $I_L' = \frac{R_N}{R_N + R_L} \times I_{sc}$

% error can be calculated as $\frac{\text{Observed value} - \text{calculated}}{\text{calculated}} \times 100$

$$= \frac{0.72 - 0.697}{0.697} \times 100$$

Result: ~~The~~ The % error is: 3.29%

Conclusion: The ~~not~~ Norton's theorem is verified and the Norton's equivalent circuit using DC source is obtained

Experiment : 3

Aim : To control a single lamp from two different places using 2-way switch circuit.

Apparatus Used : Incandescent lamp, 2-way switch kit & connecting cables.

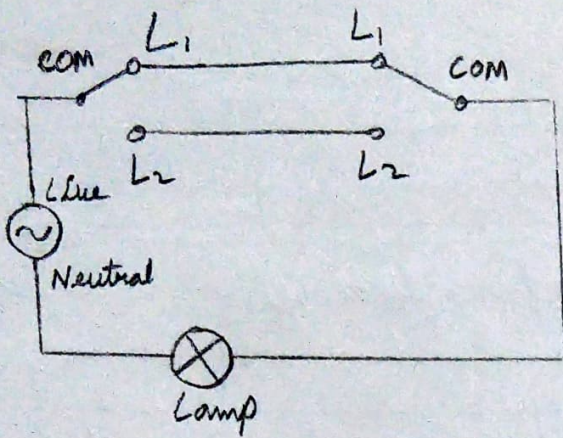
Theory : A two way switch is installed near the first step of the stairs. The other 2 way switch is installed at the upper part, where the stairs end.

The light point is provided b/w the first and the last stairs at an adequate location and height, if the light is switched on by the switch at the bottom, it can be turned off by the switch at the top or vice-versa. The circuit can be used at places like bedroom, where the person may not have to move from their place to turn the switch off. They can conveniently control the light source from either of the switches.

Two switches are used in a 2-way switch circuit. The supply is given to the switch at the short-circuited terminals. The connection to the light point is taken from the similar short-circuited terminal of the second switch. Other two independent terminals of each circuit are connected through cables.

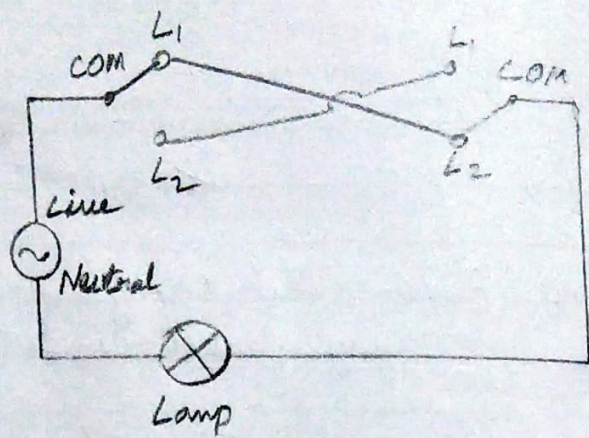
Procedure : → While the supply and the 2-way switch kit are turned off, connect the bulb into the holder of the kit.

→ Connected the terminals of both the switches S1 & S2 for direct connection.



DIRECT CONNECTION

(Both switches is same condition)



CROSS CONNECTION

(Both switches is opposite condition)

DIRECT CONNECTION			CROSS CONNECTION		
POSITION OF SWITCHES		CONDITION OF LAMP	POSITION OF SWITCHES		CONDITION OF LAMP
S1	S2		S1	S2	
ON	ON	ON	ON	ON	OFF
ON	OFF	OFF	ON	OFF	ON
OFF	ON	OFF	OFF	ON	ON
OFF	OFF	ON	OFF	OFF	OFF

→ Switch in the supply and the Kit. observe it and note the status of the bulb.

→ Turn off the Kit and the power supply. Repeat step 2 & 3 for Cross connection as per the circuit diagram.

Result: In Connection the lamp was turned on when both the switches were in the same position.

In Connection the lamp was turned on when both the switches were in ~~different~~ opposite positions.

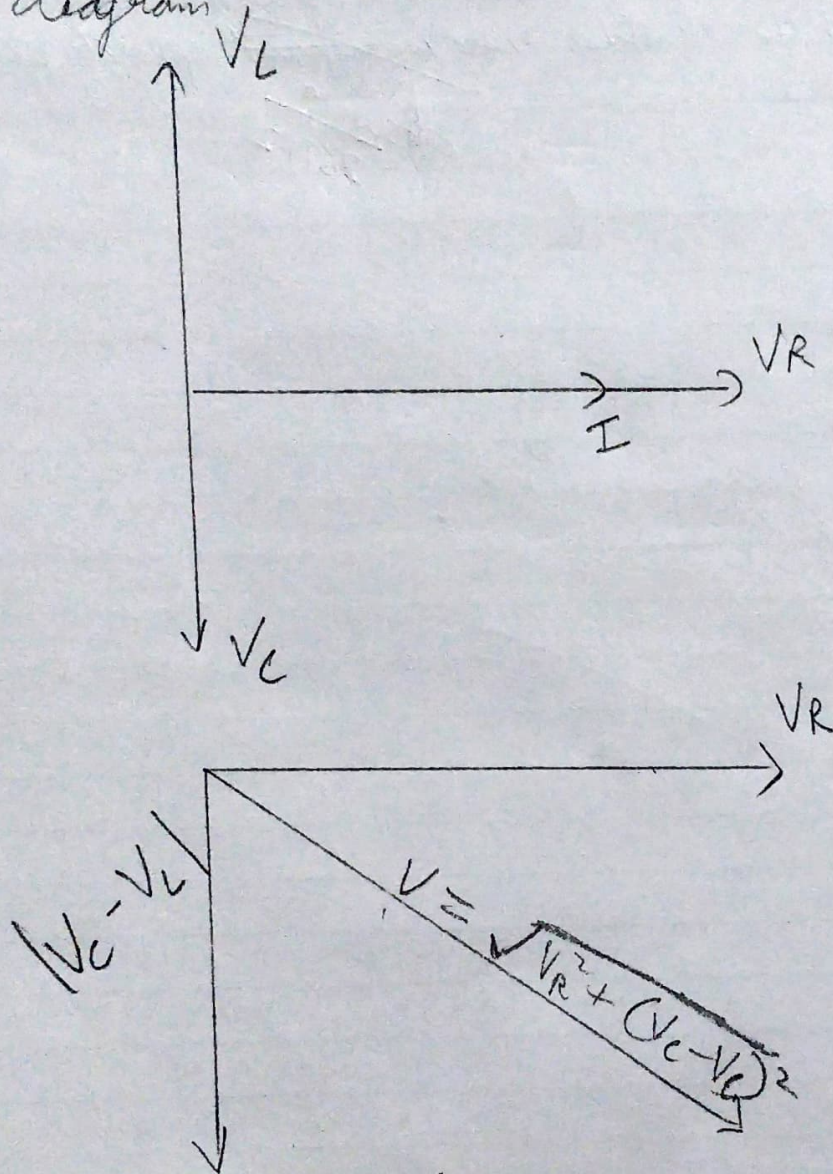
Ashwin
26/12

Observation:

V_R	V_L	V_C	I_C
79.5V	163.1V	371V	0.26A

Phase diagram

1.



Experiment - 4

Aim: To study R-L-C series circuit and draw its phasor diagram.

Apparatus: Ammeter (0-1A), resistor, inductor and capacitor, connecting wires and digital multimeter

Theory: Considering an AC circuit containing resistance R , inductance L H and capacitance C F in series.

Let I A current be flowing through the circuit and supply frequency of f Hz.

Now, Voltage drop across:

(a) Inductance, $V_L = I(\omega L)$, V leading by 90°

(b) Capacitance, $V_C = I\left(\frac{1}{\omega C}\right)$, I leading by 90°

(c) Resistance, $V_R = IR$, In phase with I

Phase angle between voltage & current:

$$\phi = \tan^{-1} \left[\frac{V_L - V_C}{V_R} \right]$$

$$\text{Power factor (PF)} = \cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\text{Power consumed in the circuit} = P = I^2 R \text{ or } VI \cos \phi$$

Calculations:

$$R = \frac{V_R}{I} = \frac{79.5}{0.26} = 286.53 \Omega$$

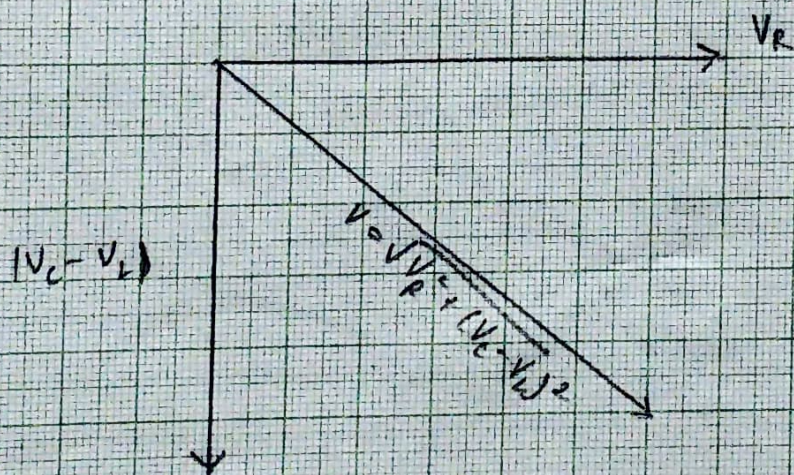
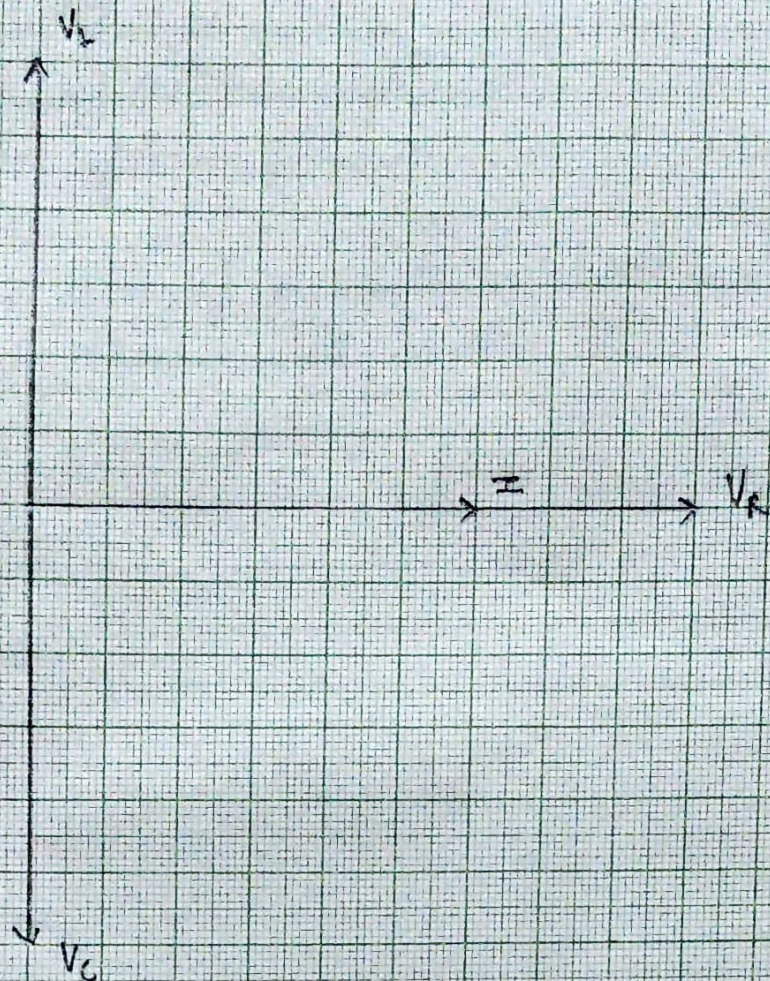
$$X_L = \frac{V_L}{I} = \frac{183.1}{0.26} = 627.307 \Omega$$

$$X_C = \frac{V_C}{I} = \frac{371}{0.26} = 1426.923 \Omega$$

$$\begin{aligned} \text{Impedance} = Z &= \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{(286.53)^2 + (1426.923 - 627.307)^2} \\ &= 898.41 \Omega \end{aligned}$$

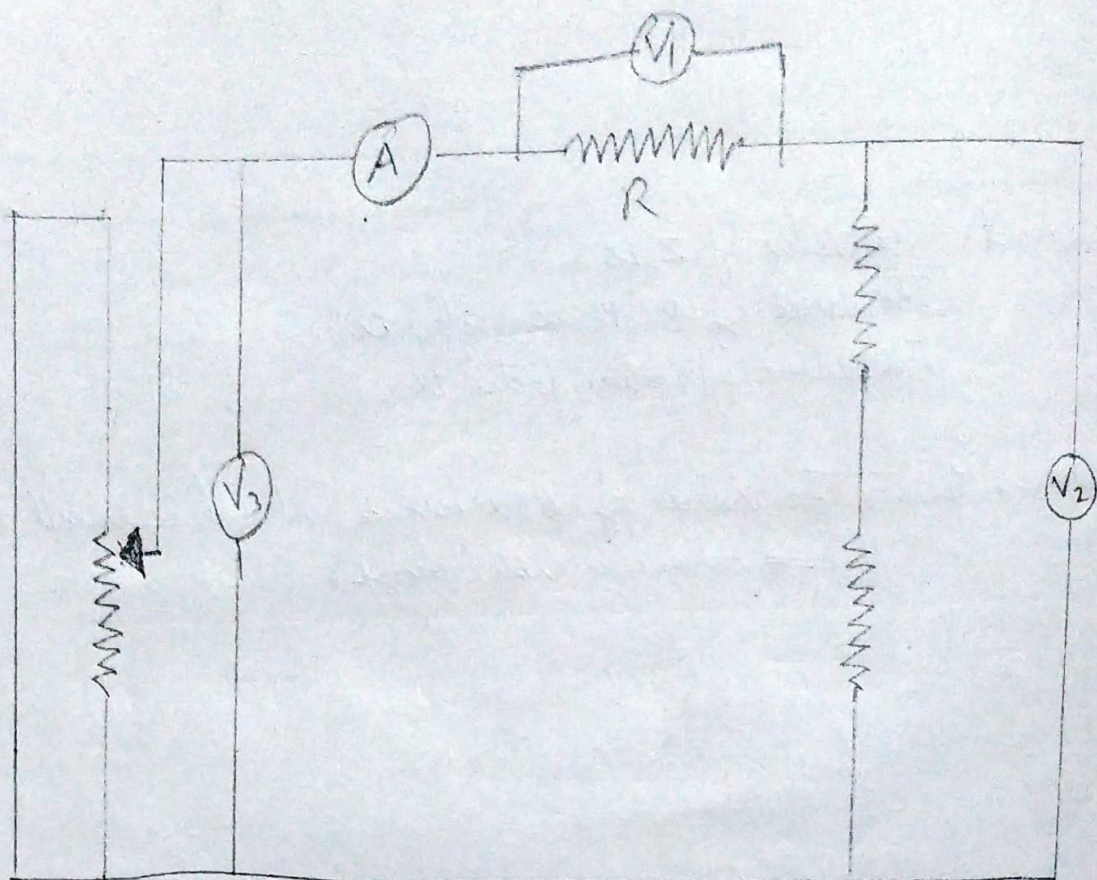
$$\begin{aligned} \text{Phase angle} &= \tan^{-1} \left(\frac{V_L - V_C}{V_R} \right) \\ &= \tan^{-1} (-279) \\ &= -70.281 \end{aligned}$$

$$\text{Power factor} = \cos \phi = 0.337$$

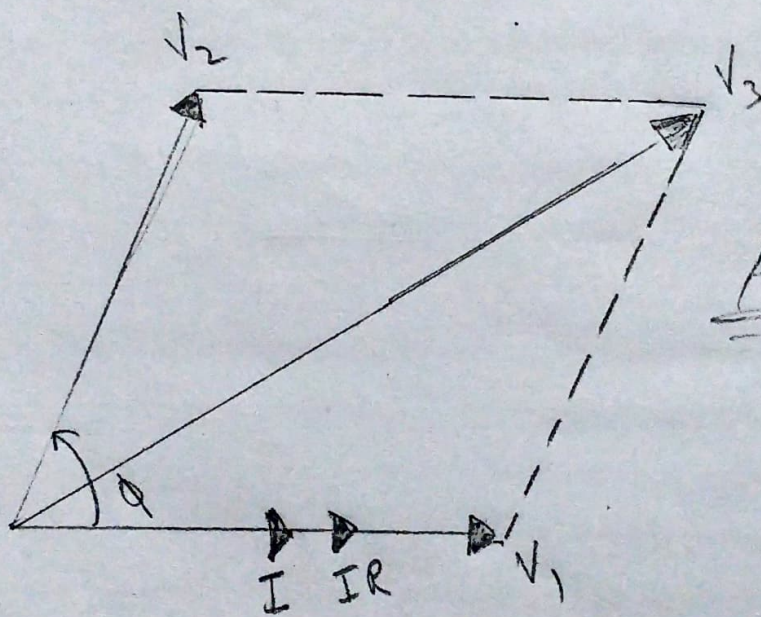


Result : The value of Z is
The value of ϕ (Phase angle) is
The value of Power factor is

Precautions : The value of ~~a~~ resistance should be small otherwise
~~the~~ resonance will occur.



Circuit Diagram



Phase Diagram

Experiment - 5

Aim: To measure the single phase power in a single phase circuit by using three voltmeters.

Apparatus: AC Wattmeter (0-300V, 10Amp), 2 AC Voltmeters (0-180V), 1 AC voltmeter (0-300V), Variac 230V 10A 50Hz, I-Phase, single phase load, rheostat & connecting wires.

Theory: Power consumed by load $\Rightarrow P = V_2 I \cos \phi$

From the phasor diagram we can ~~say~~ write,

$$V_3^2 = V_1^2 + V_2^2 + 2V_1V_2 \cos \phi$$

Power factor, $\cos \phi = \frac{(V_3^2 - V_1^2 - V_2^2)}{2V_1V_2}$

$$I = \frac{V_1}{R} \quad [\text{here } R = \Omega]$$

Now, $P_{\text{calculated}} = V_2 I \cos \phi$

$$= V_2 \left(\frac{V_1}{R} \right) \cos \phi$$

$$= \left[\frac{V_1 V_2}{R} \right] \left[\frac{V_3^2 - V_1^2 - V_2^2}{2V_1 V_2} \right]$$

$$= \frac{1}{2R} [V_3^2 - V_1^2 - V_2^2]$$

From the above eqⁿ, it can be observed that, the power and power factor in an AC circuit can be measured by using 3-single phase voltmeter, instead of a wattmeter.

SNo	V_1	V_2	V_3	P_{calc}	Wattmeter	$\cos \phi$
1	30V	92V	100V	7.93W	7.5W	0.115

$$\text{Percentage error} = \frac{P_{\text{calculated}} - \text{Wattmeter Reading}}{\text{Wattmeter Reading}} \times 100$$

- Procedure:
- ① Make the connections as per the circuit diagram
 - ② Keep the variac at zero position before starting the experiment.
 - ③ Switch on AC supply
 - ④ By Varying the variac set the Voltmeter ~~read~~ reading at supply voltage.
 - ⑤ Vary the R_L load to obtain different readings of voltmeter and wattmeter
 - ⑥ Repeat step 5 for different observations.
 - ⑦ Set the Variac at zero position and switch off supply

Calculations:

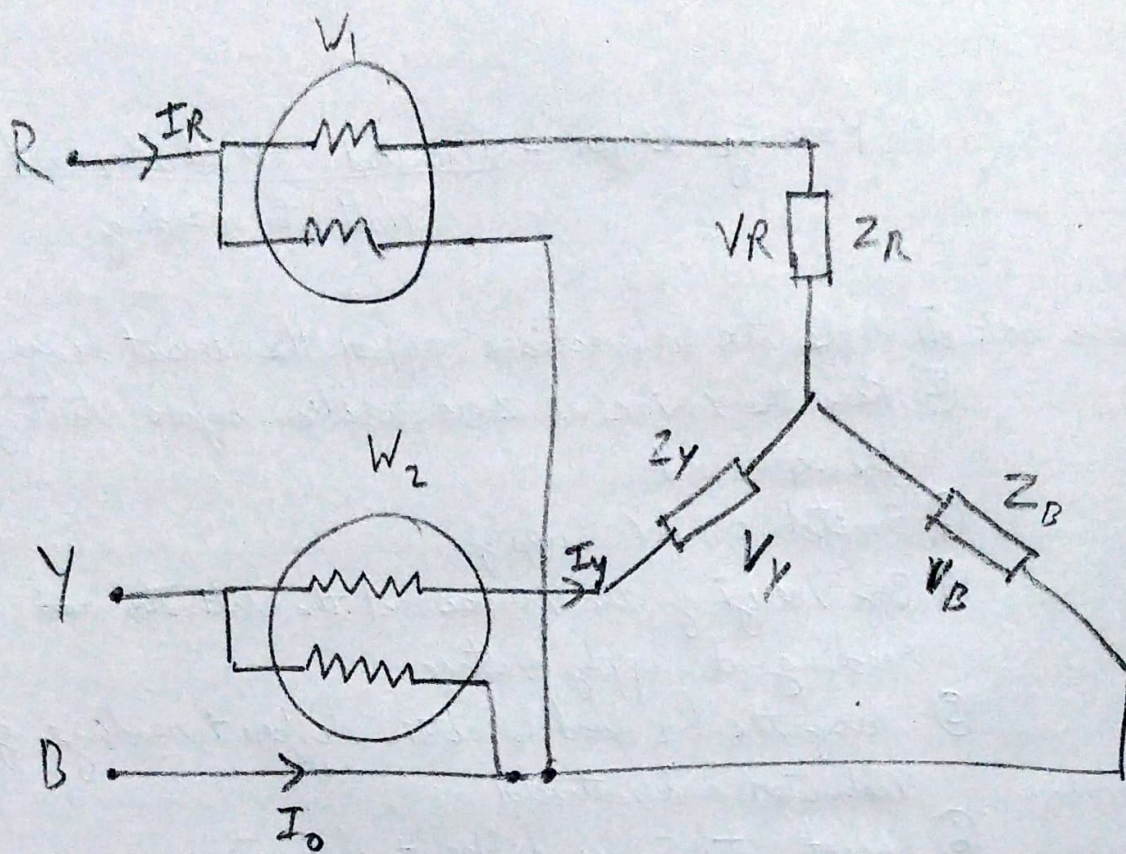
$$\text{Percentage Error} = \frac{P_{\text{calc.}} - \text{Wattmeter}}{\text{Wattmeter}} \times 100$$

$$\% \text{ Error} = \frac{7.93 - 7.5}{7.93} \times 100$$

$$\% \text{ Error} = \frac{0.43}{7.93} \times 100$$

$$\% \text{ Error} = 5.73\%$$

Result: The percentage errors were found to be 5.73%.



Circuit diagram

Experiment - 6

Aim:- To measure power and power factor of a balanced three phase star-connected load using two wattmeter method.

Apparatus:- 2 Wattmeter (5/10 A, 150/300/500 V), Voltmeter (0-500 V), Ammeter (0-10 A) & 3 phase load (Resistive).

Theory:-

Power consumed by a 3-phase balanced star connected load can be measured by using two wattmeters connected in the circuit. The current coils of the wattmeters are connected in series with the load in any two lines and the pressure coils are connected b/w these lines. The current through the coil of W_1 is I_R and the current through the current coil of W_2 is I_B . The potential difference across the pressure (voltage) coil of W_1 is $V_{RY} = V_R - V_Y = V_{RY}$. The phase difference between I_R and V_{RY} is $(30^\circ + \phi)$, while that between I_B and V_{BY} is $(30^\circ - \phi)$. Hence, the reading of W_1 is given as $W_1 = V_{RY} I_R \cos(30^\circ - \phi)$.

Similarly, the reading of W_2 is given as $W_2 = V_{BY} I_B \cos(30^\circ - \phi)$. Since the load is balanced, therefore $|I_R| = |I_Y| = |I_B| = I$ and $|V_{RY}| = |V_{BY}| = V_L$. Thus, $W_1 = V_L I \cos(30^\circ + \phi)$ and $W_2 = V_L I \cos(30^\circ - \phi)$. Then the total power is given as:

$$P = W_1 + W_2 = V_L I \cos(30^\circ + \phi) + V_L I \cos(30^\circ - \phi) = \sqrt{3} V_L I \cos \phi$$

also, $W_2 - W_1 = V_L I \sin \phi$

$$\text{Therefore, } \tan \phi = \sqrt{3} \frac{W_2 - W_1}{W_2 + W_1}$$

Observations and Calculations :-

$W_1 (W)$	$W_2 (W)$	$\cos \phi$
1040	1040	1

$$W_1 = W_2 = 1040 \text{ Watts}$$

$$\phi = \tan^{-1} \sqrt{3} \frac{(W_1 - W_2)}{(W_1 + W_2)}$$

$$\phi = \tan^{-1} \sqrt{3} \frac{(1040 - 1040)}{(1040 + 1040)}$$

$$\phi = \tan^{-1}(0)$$

$$\phi = 0$$

$$\boxed{\cos \phi = \cos 0 = 1}$$

Procedure :-

1. Connect the resistive load and the two wattmeters externally as per the diagram. For the lower current ranges, short the terminals B1 and B2 of the wattmeter. For higher current range, short the terminals B1-E1 and B2-E2. Ensure that the three-phase load is off-position.
2. Switch on the three-phase AC mains and increase the applied voltage till rated value 415V (as indicated by the voltmeter) is achieved.
3. Switch on the load and turn on the resistive load for various load settings, as indicated by the ammeter till its rated load current.
4. Note the readings of the wattmeter for various load settings.
5. Decrease the load gradually, turn off the load and switch-off the supply.
6. Note the multiplication factor of the wattmeter.

Result :- Total power in the three phase balanced star-connected load was found to be 2080Watts, and the power factor was calculated as 1.

Experiment - 7

Aim:- To perform open circuit and short circuit test on a single phase transformer.

Apparatus:- 2 AC Wattmeters [0-75W & 0.250W],
 2 AC Voltmeter [0-300V & 0-250V],
 2 AC Ammeter [0-25A & 0-2.5A],
 2 Variac [230V, 10A, 50Hz, 1-phase] & [230V, 10A, 50Hz, 1-phase],
 Transformer [1 phase & 50Hz] & Connecting Wires.

Theory:- For open circuit $W = V_1 I_0 \cos \phi_0$; $\cos \phi_0 = \frac{W}{V_1 I_0}$

$$I_p = I_0 \sin \phi_0, I_w = I_0 \cos \phi_0$$

$$X_0 = \frac{V_1}{I_p}, R_0 = \frac{V_1}{I_w}$$

$$I_0 = V_1 y_0, y_0 = \frac{I_0}{V_1} \quad W = V_1^2 G_0; G_0 = \frac{W}{V_1^2}$$

$$B_0 = \sqrt{y_0^2 - G_0^2}$$

For short circuit

$$R_1 = \frac{V_{sc}}{I_{sc}^2}, Z_1 = \frac{V_{sc}}{I_{sc}}$$

$$X_1 = \sqrt{Z_1^2 - R_1^2}$$

Observations for open circuit :-

V_o	I_o	W_o
114V	0.02A	2W

Observations for closed circuit :-

V_{sc}	I_{sc}	W_o
20V	2.39A	48W

Calculations :-

$$\cos \phi = \frac{W_o}{V_o I_o} = \frac{2}{114 \times 0.02} = 0.87$$

$$\eta = \frac{\text{output} \times \cos \phi \times 100}{\text{output} \times \cos \phi \times \text{losses}} = \frac{870 \times 100}{870 \times 50}$$

$\eta = 94.56\%$

Procedure:-

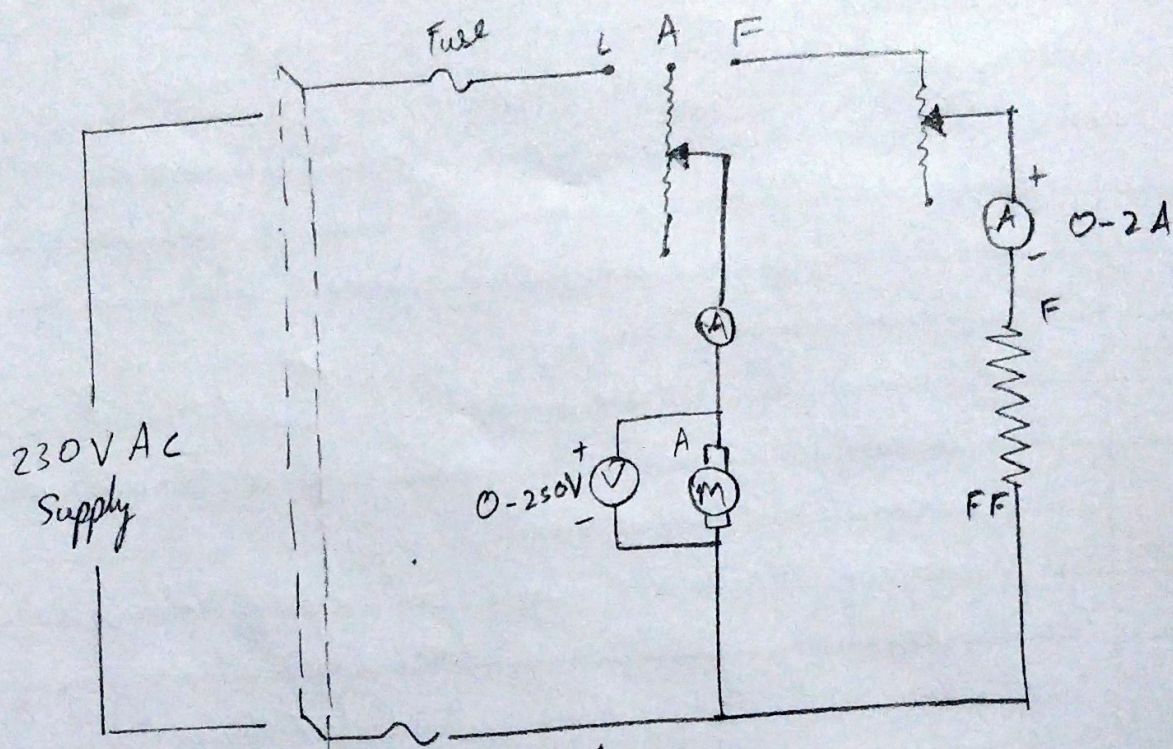
1. Make the connections as per the circuit diagram.
 2. Make sure that the secondary side of transformer is open.
 3. Keep the Variac at zero position before switching on the supply.
 4. Switch on the AC supply.
 5. By varying the Variac, apply full load current to the transformer and note the reading of voltmeter, wattmeter and ammeter (Open circuit).
- By varying the Variac apply full supply voltage i.e. 230V to the primary of the transformer and note the reading of voltmeter, wattmeter and ammeter (Short circuit).
6. Keep the Variac at 0 position (Zero) and switch off supply.

Result:-

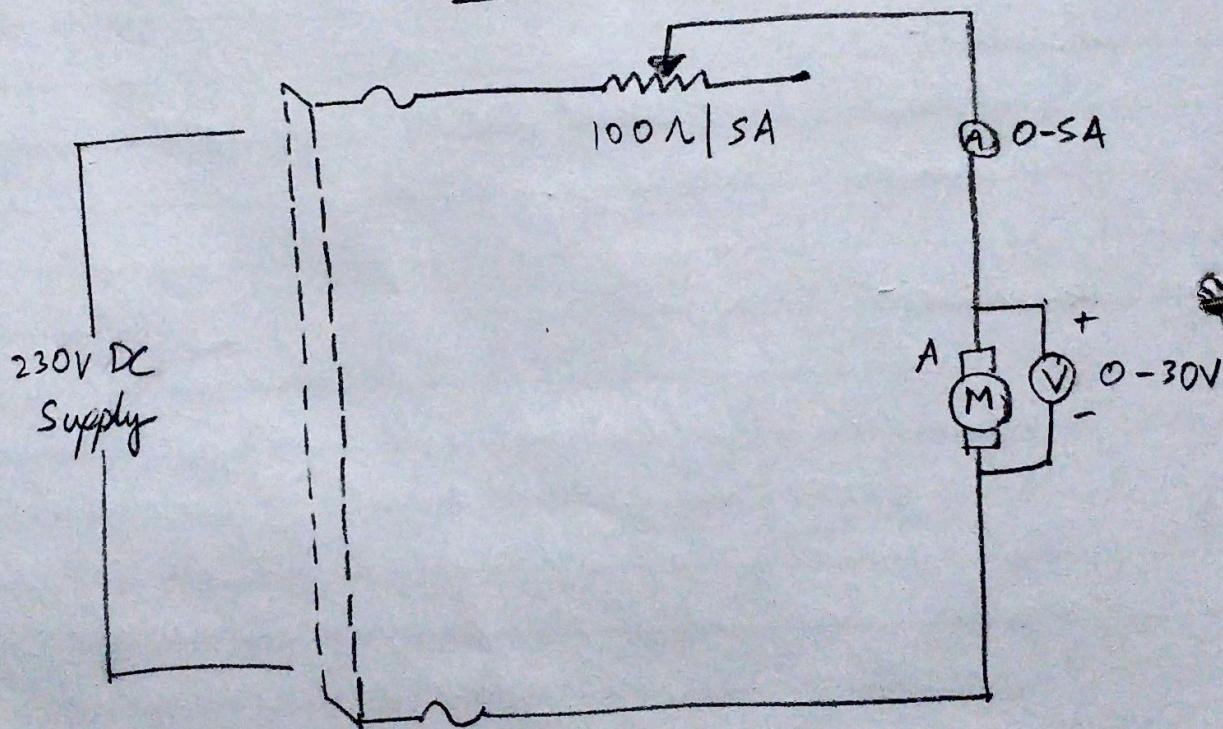
Open circuit & short circuit test on single-phase transformer have been done successfully.

Precautions:-

1. All the connections should be tight and clean.
2. Special care should be taken while selecting the ranges of the meters for conducting short circuit and open circuit tests.
3. While conducting the short-circuit test, the voltage applied should be initially set at zero, and then increase slowly. If a little higher voltage than the required voltage be applied (by mistake), there is a danger of transformer being damaged.



Circuit diagram



Armature Resistance

Experiment - 8

Aim :- To study speed control of DC shunt motor

Apparatus :- Ammeter [0-5A] & [0-2A], Voltmeter [0-250V], Rheostats [100/5A] and [400/1.7A], Tachometer [0-2000 rpm] and connecting wires

Theory :- i) Armature Voltage control Method

For a load of constant torque, the speed is proportional to the applied voltage across armature. Therefore speed voltage characteristic is linear and a characteristic is a type of straight line. As the voltage decreases across the armature, the speed falls. This method gives speeds less than ~~rotated~~ rated speeds.

$$E_b \propto \phi N$$

$$E_b \propto N$$

$$V - I_a (R_a + R) \propto N$$

As the voltage decreases speed increases.

ii) Flux Control Method

With rated voltage applied to the motor the field resistance is increased i.e. field current is decreased. It is observed that speed increases.

$$\frac{E_b}{\phi} \propto N \quad \text{or} \quad N_b \propto \frac{E_b}{I_f}$$

The characteristics I_f vs N is inverse and represented by a hyperbola.

Observation :-

S No.	Flux control Method		Armature Constant Method	
	Field current	Speed	Armature Voltage	Speed
1	0.35	1400	70	1400
2	0.40	1300	62	1250
3	0.45	1294	54	1036

Procedure:- i) Armature Voltage Control Method

1. Make connections as per the circuit diagram
2. Show the connections to the lab instructor
3. Keeping both rheostats at minimum, start the motor with the help of a starter and by adjusting field rheostat bring the motor to rated speed.
4. By increasing armature circuit rheostat in steps, note down the voltage, I_a and speed at every step.
5. The corresponding graph is drawn between armature voltage V_a vs. speed.

ii) Flux Control Method

1. Measure, The machine run at its rated speed and rated voltage obtained.
2. The voltage is kept constant and for different values of field current, the speed is noted.

Result :-

Speed control methods of DC shunt motor have been studied successfully.

Precautions :-

Make ~~sh~~ sure that your connections are correct.

Do not touch the live wire.

Take observations carefully.

Experiment Kit should be properly grounded.